Quantum Mirror™

Powered by Genesis OS

Premium Relationship Intelligence Assistant - Master Product, Data, AI, and Engineering Blueprint

Build-ready specification for subscription chat, memory, structured extraction, guidance, moderation, consent, storage, and cost-controlled deployment

Purpose: Define the complete blueprint for Quantum Mirror as a premium, context-aware AI experience that reads Genesis OS data, asks better questions, extracts structured signals from narrative history, and returns evidence-banded guidance without drifting into therapy, diagnosis, or unsafe overreach.

Primary design rule: The user experiences elegant reflection. The system underneath is deterministic, versioned, consent-gated, cost-controlled, and auditable.

Audience: Founders, product, backend, mobile, AI/ML, data, analytics, trust & safety, QA, and future licensing partners.

Document Map

1. Executive concept and product positioning
2. What Quantum Mirror is and is not
3. Premium experience design and subscription model
4. User journeys and conversation modes
5. Data capture model: what the system asks, stores, labels, and ignores
6. Canonical taxonomy for words, tone, phrasing, themes, and behavioral signals
7. Core database schema and field-level definitions
8. Session memory, extraction, and structured profile mutation rules
9. AI orchestration, prompt stack, evidence bands, and refusal logic
10. API contracts, service boundaries, and event model
11. Moderation, escalation, and consent architecture
12. Cost control, rate limits, and monetization logic
13. Narrow-v1 launch path, QA, analytics, and rollout gates
14. Appendices: canonical enums, sample payloads, extraction heuristics, and implementation checklists

1. Executive Concept and Product Positioning

Quantum Mirror is not a generic chatbot bolted onto Relationship Scores. It is the premium private intelligence layer of Genesis OS. The user enters a guided clarity session that can explain scores, decode emotional patterns, prepare difficult messages, map trigger loops, interpret dates or conflicts, and progressively deepen the user's emotional blueprint through structured extraction from conversation.

The premium value is not raw AI access. The premium value is context-aware guidance grounded in the user's assessment data, longitudinal behavior, reflection history, emotional state logs, attachment triggers, and evidence-banded forecast signals.

Business rule: Quantum Mirror should be framed and priced as access to a private intelligence suite. Behind the scenes, usage controls, memory windows, evidence thresholds, and storage policies protect both margin and trust.

Core product promise

- Help users understand themselves more clearly than a static score can.
- Translate confusion into structured insight without pretending to be therapy or clinical diagnosis.
- Capture deeper, narrative-grade emotional data that multiple-choice questions will never fully surface.
- Turn reflective conversations into governed signals that improve readiness, matching, forecasting, coaching, and retention.
- Monetize recurring value in a way that naturally covers AI compute costs.

What the user feels

- Private access: the feature should feel reserved, intelligent, and personal.
- Continuity: the assistant remembers prior sessions appropriately and builds on them.
- Precision: the assistant references the user's scores, patterns, and history with calm specificity.
- Respect: the assistant does not pry for trauma theater or push users to disclose more than they want.
- Safety: the assistant is helpful inside guardrails and knows when to step back or escalate.

2. What Quantum Mirror Is and Is Not

It is

- A context-aware reflection and guidance assistant powered by Genesis OS data.
- A premium layer for score interpretation, pattern translation, communication help, post-date reflection, post-conflict decoding, and guided self-awareness.
- A structured data collection surface that can ask targeted questions about past relationships, trust ruptures, conflict loops, values, fears, hopes, and growth.
- A governed insight engine that converts narrative conversation into canonical tags, metrics, and evidence-banded suggestions.

It is not

- Not a replacement for therapy, emergency support, legal advice, medical advice, or clinical diagnosis.

- Not a truth oracle about a partner who has not consented to data sharing.
- Not a freeform memory bucket that stores everything forever.
- Not a place where AI can mutate core profile truth on weak evidence.
- Not an unmetered cost sink disguised as premium.

Hard non-negotiables

- Deterministic first: visible scores and durable profile states must trace back to versioned formulas, thresholds, and canonical fields.
- Consent-gated memory: no derived partner-facing insight may be produced from non-consented private data.
- Evidence bands: every strong statement must carry an internal evidence level and confidence basis.
- Separation of layers: slow identity, medium-horizon patterning, and fast emotional state must not be collapsed into one blob.
- Auditable extraction: any structured signal saved from chat must store provenance, evidence snippets, extractor version, and review status.

3. Premium Experience Design and Subscription Model

Recommended tiering

Tier	User sees	System limits	Primary goal
Free preview	Read score explanation, a few guided prompts, one light mirror check, and upsell surfaces	Small context window, no deep memory, no narrative extraction persistence, low-cost model path	Create curiosity and trust
Quantum Mirror Premium	Full personal sessions, pattern decoding, message drafting, reflection analysis, guided past-relationship intake, state check-ins, growth plans	Moderate monthly message/session cap, full personal memory scope, governed extraction, standard model path	Primary subscription revenue
Quantum Mirror Couple	Both users consent; couple dynamic reflection, repair prompts, shared review tools, evidence-banded joint summaries	Requires dual consent, stricter moderation, shared-memory rules, capped co-analysis depth	Higher-value relationship mode later
Enterprise / Licensed API	White-labeled or partner access under contract	Contractual rate limits, no direct user memory by default, isolated tenant store	Licensing and B2B expansion

Premium hooks and UI framing

- Use premium entry language such as "Enter your private clarity session", "Unlock deeper insight from your Genesis OS blueprint", and "Ask what the score alone cannot explain."
- Preview cards should show restrained mystery: a blurred sample insight, a lock icon, and copy that promises pattern-aware guidance rather than generic chat.
- Display category chips such as Pattern Translation, Message Help, Past Relationship Decode, Trigger Mapping, Score Deep Dive, and After the Date.
- Show a subtle continuity note: 'Built from your assessment, logs, and prior sessions.' This communicates personalization without sounding creepy.
- Display a premium ledger screen summarizing what Quantum Mirror has learned with explicit user controls over saved memories and extracted signals.

4. User Journeys and Conversation Modes

Primary conversation modes

Mode	Why user enters	Inputs read	Outputs returned
Score Deep Dive	User asks what a score, archetype, shadow pattern, or match score means.	readiness score, compatibility score, archetype, shadow, evidence bands, trend history	explanation, risk areas, next steps, links to reflections
Past Relationship Decode	AI interviews user about a past relationship to collect narrative data and lessons.	relationship history intake, trigger themes, betrayal patterns, ending style, grief residue	structured history extract, empathy summary, growth signals
Message Studio	User wants help wording or interpreting a message.	current state, match context, trigger flags, communication style, evidence band	draft suggestions, tone options, caution notes
After the Date / After the Conflict	User wants to interpret recent interaction.	event log, emotional state check-in, perception-vs-reality markers, trigger history	grounded reflection, follow-up questions, state update
Pattern Check	User asks why the same thing keeps happening.	longitudinal trends, reflection clusters, repeated tags, past relationship themes	pattern summary, evidence basis, intervention ideas
Couple Reflection	Both parties consent to shared review.	dual profiles, shared events, state deltas, consent scope, evidence band	joint summary, repair prompts, no unilateral exposure

Guided intake flows the AI should initiate

Past relationship intake

- Which relationship changed you the most, and what made it leave a mark?
- What behavior in that relationship made you feel most unsafe, small, or confused?
- During conflict, did you move closer, get quiet, get sharp, or emotionally check out?
- What did you keep excusing longer than you should have?
- What part of healthy love felt unfamiliar or harder to trust?

Trigger mapping

- What kind of silence hits you hardest - delayed texts, sudden distance, cold tone, or mixed signals?
- What makes your body feel like it needs to protect itself fast?
- When you feel cornered emotionally, do you shut down, explain harder, leave, or perform calm?

Growth check-in

- What did you handle better this time than you would have six months ago?
- What did you feel proud of, and what still felt old or automatic?

5. Data Capture Model: What the System Asks, Stores, Labels, and Ignores

Quantum Mirror should treat data capture as a governed funnel. Not every utterance becomes memory. Not every memory becomes a canonical field. Not every canonical field can influence scoring.

Three-layer data funnel

- Layer A - ephemeral session context: recent user messages, current intent, visible scores, last few state logs. Used for response generation; may expire quickly.
- Layer B - session memory artifacts: notable facts, preferences, or narrative anchors that help future continuity. Stored only if within consent and product value scope.
- Layer C - structured profile signals: canonical tags, metrics, history objects, and counts that can influence analytics, scoring, or future AI. Saved only when evidence threshold and schema rules are met.

Data classes Quantum Mirror may collect

Data class	Examples	Saved as
Personal reflection data	How the user describes feelings, hopes, fears, regrets, priorities, and growth.	reflection summary, values tags, need tags, concern tags
Relationship history data	Past partner dynamics, betrayals, exits, conflict style, recurring pain themes.	history entries, rupture tags, ending style, unresolved theme tags

Data class	Examples	Saved as
Live interaction interpretation	How a recent conversation or date felt and what objectively occurred.	state check-ins, event links, perception-vs-reality gap metrics
Communication preferences	What wording style, reassurance style, pace, and tone the user responds to.	preferred coaching tone, communication comfort tags
Risk/safety signals	Harassment, coercion, self-harm concern, stalking, threats, panic, abuse cues.	moderation events, urgency class, escalation path
Meta usage signals	What modes they use, session length, helpfulness score, conversions, drop-offs.	analytics events, subscription metrics, cost metrics

Data Quantum Mirror should explicitly avoid saving as durable memory unless separately justified

- Incidental gossip about unrelated third parties.
- Sensitive trauma details that do not improve product value or safety handling.
- Speculative diagnoses about self or others.
- High-risk content that belongs in moderation evidence storage rather than product memory.
- One-off emotional spikes that have not repeated and do not meet extraction thresholds.
- Anything outside the user's consent scope or retention setting.

6. Canonical Taxonomy for Words, Tone, Phrasing, Themes, and Behavioral Signals

Words should not be stored as fuzzy vibes. Quantum Mirror needs a controlled taxonomy that maps user language into canonical labels. This allows analytics, AI, and scoring to speak the same internal language.

6.1 Taxonomy layers

- Surface Lexicon Layer: exact words or phrases spoken by the user.
- Interpretation Layer: what those words likely indicate, stored as controlled tags.
- Behavior Layer: inferred action pattern such as pursuit, withdrawal, soft entry, overexplaining, appeasing, or defensive sharpness.
- State Layer: present emotional state such as guarded, activated, ashamed, hopeful, drained, or numb.
- Narrative Theme Layer: broader recurring themes such as abandonment fear, emotional invisibility, inconsistency addiction, rescue role, loyalty-over-self, or peacekeeping at own expense.

6.2 Word / tone / phrasing taxonomy model

Source layer	Stored field	Example raw phrasing	Canonical label example	Extraction rule	Used for
Word choice	lexical_markers[]	I keep rereading every text	LEX_REREADS_MESSAGES	literal or near-literal phrase match	supporting evidence only
Tone framing	tone_profile.primary	Depends on the day honestly...	TONE_HEDGED_REFLECTIVE	hedge + reflective wording pattern	response style and extraction confidence
Self-positioning	self_positioning_tag	I usually make myself smaller	SELF_MINIMIZING	first-person self-reduction cue	readiness coaching
Conflict phrasing	conflict_posture_tag	I joke first, then maybe say it	CONFLICT_SOFT_ENTRY	humor-as-entry + delayed directness	communication style
Attachment cue	attachment_trigger_tag	late replies make my chest drop	TRIGGER_DELAYED_RESPONSE	body alarm + delay reference	trigger mapping
Narrative theme	history_theme_tags[]	I keep staying too long	THEME_OVERSTAYING	repetition + regret framing	past relationship analysis
Boundary signal	boundary_expression_tag	I do not want to beg to be chosen	BOUNDARY_DIGNITY_PRESERVE	self-protection statement	growth signal
Behavior inference	behavior_pattern_tag	I disappear before they can switch up	PATTERN_PREEMPTIVE_WITHDRAWAL	leave-first protection logic	forecasting
Repair cue	repair_style_tag	I come back when I cool off	REPAIR_DELAYED_REENGAGE	distance then reconnect	couple guidance
Risk cue	safety_flag	He kept showing up after I said stop	SAFETY_STALKING_CONCERN	clear coercive persistence	moderation escalation

6.3 Controlled enums for tone, phrasing, and narrative posture

ToneProfile.primary: TONE_DIRECT, TONE_HEDGED_REFLECTIVE, TONE_WARM_OPEN, TONE_DRY_DETACHED, TONE_GUARDED, TONE_OVEREXPLAINING, TONE_CHAOTIC_ACTIVATED, TONE_FLAT_EXHAUSTED, TONE_SELF_BLAMING, TONE_ACCUSATORY

PhrasingStyle.tags: PHRASE_JOKING_DEFLECTION, PHRASE_QUALIFIER_HEAVY, PHRASE_MIND_READING, PHRASE_ABSOLUTE_LANGUAGE, PHRASE_BODY_SIGNAL_REFERENCE, PHRASE_PAST_PATTERN_REFERENCE, PHRASE_GROWTH_OWNERSHIP, PHRASE_PEACEKEEPING, PHRASE_CONTROLLED_DISTANCE, PHRASE_REASSURANCE_SEEKING

NarrativeTheme.tags: THEME_ABANDONMENT_FEAR, THEME_EMOTIONAL_INVISIBILITY, THEME_INCONSISTENCY_ADDICTION, THEME_OVERFUNCTIONING, THEME_SELF_ERASURE, THEME_LOYALTY_TO_WOUND, THEME_FIXING_OTHERS, THEME_SAFETY_CRAVING, THEME_CONTROL_AVOIDANCE, THEME_TRUST_FRACTURE, THEME_GRIEF_RESIDUE

BehaviorPattern.tags: PATTERN_PURSUER, PATTERN_DISTANCER, PATTERN_PASSIVE_DRIFTER, PATTERN_SOFT_ENTRY, PATTERN_HYPER_VIGILANT_DECODER, PATTERN_PREEMPTIVE_WITHDRAWAL, PATTERN_OVEREXPLAINER, PATTERN_APPEASER, PATTERN_SCOREKEEPER, PATTERN_REPAIR_ORIENTED

6.4 Labeling method

- Store the raw phrase only in session evidence or moderated memory, not as the durable analytic label.
- Map raw language to canonical labels using extractor rules plus confidence scoring.
- Allow multiple labels per utterance when supported; do not force a single-label narrative if the user shows mixed posture.
- Preserve negation and time context. 'I used to overexplain' should not be stored the same as 'I overexplain now.'
- Maintain label provenance: every stored tag must point back to source message id, source span, extractor version, and save reason.

7. Core Database Schema and Field-Level Definitions

Quantum Mirror requires its own bounded schema family while reusing Genesis OS canonical identity, scoring, state, and event objects. The system should not hide core logic in prompts; it should expose it as database truth.

7.1 quantum_mirror_subscription

Field	Type	Required	Description	Notes / Rules
subscription_id	uuid	yes	Primary subscription record	One active row per user per product tier
user_id	uuid	yes	Links to user profile	FK users.user_id
tier_code	enum	yes	FREE_PREVIEW / QM_PREMIUM / QM_COUPLE	Do not use free-text tier names
billing_provider	enum	yes	APPLE / GOOGLE / STRIPE / PARTNER	Needed for entitlement sync
billing_status	enum	yes	ACTIVE / PAST_DUE / CANCELED / TRIAL / PAUSED	Source of truth for entitlement
period_start_at	timestamp	yes	Billing cycle start	UTC
period_end_at	timestamp	yes	Billing cycle end	UTC
session_quota	int	yes	Allowed premium sessions in period	Can be null for unlimited plan if cost model permits
message_quota	int	yes	Allowed messages in	Soft or hard cap

Field	Type	Required	Description	Notes / Rules
			period	
context_depth_tier	enum	yes	SHALLOW / STANDARD / DEEP	Controls memory and model path
created_at	timestamp	yes	Audit timestamp	UTC
updated_at	timestamp	yes	Audit timestamp	UTC

7.2 quantum_mirror_session

Field	Type	Required	Description	Notes / Rules
qm_session_id	uuid	yes	Primary session id	Created each time user enters Quantum Mirror
user_id	uuid	yes	Session owner	FK users.user_id
conversation_mode	enum	yes	SCORE_DEEP_DIVE / PAST_RELATIONSHIP / MESSAGE_STUDIO / AFTER_CONFLICT / PATTERN_CHECK / COUPLE_REFLECTION	Drives prompt path and extraction recipe
partner_context_user_id	uuid	no	Optional partner/match context	Only if allowed by consent scope
entry_surface	enum	yes	PAYWALL / PROFILE / MATCH / SANCTUARY / SCORE_RESULT / PUSH	Analytics and product optimization
session_status	enum	yes	OPEN / CLOSED / ESCALATED / INTERRUPTED / RATE_LIMITED	Lifecycle control
memory_scope	enum	yes	EPHEMERAL_ONLY / PERSONAL_MEMORY / PERSONAL_PLUS_COUPLE	Must obey consent
model_path	enum	yes	LITE / STANDARD / DEEP_REASONING	Cost and performance tracking
evidence_band_ceiling	enum	yes	LOW / MEDIUM / HIGH	Prevents overclaiming in low-data contexts

Field	Type	Required	Description	Notes / Rules
started_at	timestamp	yes	Session start	UTC
ended_at	timestamp	no	Session end	UTC
token_input_count	int	no	Billing/observability	Store per session
token_output_count	int	no	Billing/observability	Store per session
cost_estimate_usd	decimal	no	Approximate model cost	Used in contribution margin reporting

7.3 quantum_mirror_message

Field	Type	Required	Description	Notes / Rules
qm_message_id	uuid	yes	Primary message id	
qm_session_id	uuid	yes	Parent session	
speaker_role	enum	yes	USER / ASSISTANT / SYSTEM / MODERATOR	
message_index	int	yes	Order in session	Monotonic per session
message_text	text	yes	Normalized message body	Retain exact user text subject to deletion policy
redaction_state	enum	yes	RAW / REDACTED / MASKED	For safety and privacy
detected_intent_primary	enum	no	Top user intent	Stored after classification
tone_profile_primary	enum	no	Primary tone tag	Stored after tone analysis
safety_risk_level	enum	no	NONE / LOW / MODERATE / HIGH / CRITICAL	Moderation pipeline
created_at	timestamp	yes	Timestamp	UTC

7.4 qm_extracted_signal

Field	Type	Required	Description	Notes / Rules
signal_id	uuid	yes	Primary extracted	

Field	Type	Required	Description	Notes / Rules
			signal row	
user_id	uuid	yes	Owner of signal	
source_qm_session_id	uuid	yes	Origin session	
source_qm_message_id	uuid	yes	Origin message	Can also point to system-generated interview prompt response
signal_family	enum	yes	TRIGGER / HISTORY_THEME / COMMUNICATION_STYLE / VALUE / BOUNDARY / RELATIONAL_NEED / GROWTH_SIGNAL / SAFETY_SIGNAL / PREFERENCE / STATE	Canonical family
signal_code	varchar	yes	Controlled enum such as THEME_SELF_ERASURE	Must exist in registry
signal_value_type	enum	yes	TAG / SCORE / BOOLEAN / TEXT_SHORT / COUNT / DATE_APPROX	
signal_value	json	yes	Typed payload	Example {tag:'THEME_SELF_ERASURE'} or {score:0.74}
time_scope	enum	yes	CURRENT / PAST_PATTERN / HISTORICAL_RELATIONSHIP / STABLE_TRAIT / UNKNOWN	Prevents tense confusion
evidence_band	enum	yes	LOW / MEDIUM / HIGH	Internal certainty
extractor_version	varchar	yes	Version of extractor recipe/prompt/model	
save_status	enum	yes	PROPOSED / APPROVED_AUTO / APPROVED_REVIEWE	Governance

Field	Type	Required	Description	Notes / Rules
			D / REJECTED / EXPIRED	
provenance_snippet	text	yes	Short quoted support span	Cap length and scrub sensitive data if needed
created_at	timestamp	yes	Audit timestamp	

7.5 relationship_history_entry

Field	Type	Required	Description	Notes / Rules
history_entry_id	uuid	yes	Primary past relationship record	
user_id	uuid	yes	Owner	
relationship_label	varchar	no	User-facing label such as 'college ex'	Avoid legal names when unnecessary
relationship_type	enum	yes	DATING / COMMITTED / MARRIAGE / SITUATIONSHIP / TALKING / OTHER	
approx_start_date	date	no	Approximate start	Allow month/year granularity
approx_end_date	date	no	Approximate end	Allow null if ongoing
most_salient_themes	json	yes	Array of canonical theme tags	Derived from intake and later sessions
conflict_posture_tag	enum	no	Primary conflict style in that relationship	
ending_style_tag	enum	no	How relationship ended	MUTUAL / GHOSTED / LEFT / WAS_LEFT / CYCLICAL / UNKNOWN
grief_residue_score	decimal	no	0.00-1.00 unresolved impact score	Do not show raw decimal to user unless formatted
growth_lessons_summary	text	no	Short reflective summary	Can be AI-written and user-editable
created_at	timestamp	yes	Audit timestamp	

7.6 qm_memory_item

Field	Type	Required	Description	Notes / Rules
memory_item_id	uuid	yes	Durable memory row	
user_id	uuid	yes	Owner	
memory_type	enum	yes	PREFERENCE / HISTORY_ANCHOR / GOAL / BOUNDARY / COMM_STYLE / IMPORTANT_CONTEXT	
memory_text	text	yes	Human-readable memory statement	User can review/delete
source_signal_ids	json	yes	Signal ids supporting this memory	No orphan memories
visibility_scope	enum	yes	PRIVATE_USER / PRIVATE_SYSTEM / COUPLE_SHARED	Must obey consent
retention_class	enum	yes	STANDARD / SHORT / REGULATED / SAFETY	Drives deletion behavior
review_state	enum	yes	AUTO / USER_CONFIRMED / USER_REJECTED / REVIEWED	
created_at	timestamp	yes	Audit timestamp	

7.7 qm_helpfulness_feedback

Field	Type	Required	Description	Notes / Rules
feedback_id	uuid	yes	Primary feedback row	
qm_session_id	uuid	yes	Session being rated	
qm_message_id	uuid	no	Specific answer rated	Optional
user_id	uuid	yes	Rater	
helpfulness_score	int	yes	1-5	
clarity_score	int	no	1-5	

Field	Type	Required	Description	Notes / Rules
felt_seen_score	int	no	1-5	Optional premium signal
too_generic_flag	bool	yes	Whether answer felt generic	
too_confident_flag	bool	yes	Whether answer overclaimed	Important calibration signal
free_text_feedback	text	no	Optional comment	
created_at	timestamp	yes	Audit timestamp	

8. Session Memory, Extraction, and Structured Profile Mutation Rules

This is the heart of the system. Quantum Mirror becomes valuable when it converts reflective conversation into governed long-term intelligence without contaminating slow identity with every passing mood.

8.1 Memory classes

- Ephemeral session memory: lasts for current session and maybe a short TTL extension. Good for continuity and immediate follow-up.
- Durable personal memory: saves stable facts or recurring patterns that will improve future sessions and product value.
- Canonical structured signal: the strictest class. Can be used by scoring, forecasting, analytics, and recommendations.
- Shared couple memory: only when both parties consent to a defined scope. Never infer shared truth from unilateral disclosures.

8.2 Save/no-save decision tree

- Question 1: Is the content within allowed consent scope? If no, do not save beyond minimal operational need.
- Question 2: Is the content useful beyond the current session? If no, keep ephemeral only.
- Question 3: Does the content map cleanly to a canonical schema or memory type? If no, keep as session text or discard.
- Question 4: Is evidence strong enough? If only a single weak mention exists, save as PROPOSED signal or do not save.
- Question 5: Is the content safety-related? If yes, route through moderation evidence storage and policy logic first.
- Question 6: Could saving this create unnecessary privacy risk relative to product value? If yes, reduce granularity or do not save.

8.3 Profile mutation policy

Layer	Examples	Who can update it	Update rule
Slow identity	Archetype, shadow distribution, baseline readiness inputs, core stable preferences	Assessment engine, governed weekly jobs, explicit user edits where allowed	Never changed directly by one Quantum Mirror session
Medium pattern layer	Trigger intensities, repeated themes, communication tendencies, grief residue trend	Batch updater plus high-evidence extraction jobs	Needs repeated evidence or corroboration
Fast state layer	Current emotional state, post-conflict activation, post-date hopefulness, current stress	Immediate session and state-check services	May update per event/session; expires or decays over time
Narrative memory	Past relationship entries, durable goals, boundaries, user-approved memories	Quantum Mirror extraction + user confirmation	Editable and deletable by user

8.4 Extraction recipe

- Step 1 - classify session intent and identify extraction recipe family.
- Step 2 - run lightweight parsers for tone, lexical markers, temporal references, and safety cues.
- Step 3 - run structured extractor prompt against bounded context and canonical registry.
- Step 4 - produce candidate signal objects with evidence snippets and confidence score.
- Step 5 - apply deterministic validator: allowed enum, time-scope validity, consent scope, evidence threshold, contradiction check, duplicate suppression.
- Step 6 - save as PROPOSED or APPROVED according to policy. High-impact identity changes always route through slower governed jobs.
- Step 7 - optionally ask the user to confirm certain durable memories in plain language. Example: 'It sounds like mixed signals tend to activate your nervous system quickly. Save that as something I should remember?'
- Step 8 - log extraction metrics for review, false positive analysis, and future tuning.

9. AI Orchestration, Prompt Stack, Evidence Bands, and Refusal Logic

Quantum Mirror should be built as a pipeline, not a single monolithic prompt. The assistant must read governed packets, produce bounded outputs, and write structured artifacts through validators.

9.1 Runtime pipeline

- Orchestrator receives session request, entitlements, conversation mode, and user context.

- Context assembler fetches the minimum necessary Genesis OS packet for the request: profile summary, recent state, relevant match or partner data if consented, past Quantum Mirror memories, and evidence-band ceilings.
- Classifier stage labels intent, emotional intensity, safety risk, and extraction opportunities.
- Response planner chooses one of the approved response templates and tool paths: answer-only, answer-plus-follow-up-question, answer-plus-extraction, answer-plus-safety-escalation, or refusal-plus-resource guidance.
- Generation stage uses a system prompt specific to the mode, plus canonical labels, plus bounded evidence summary instead of raw database dumps.
- Post-processor applies safety filters, claim-strength limiter, tone normalization, and output formatting rules.
- Extractor stage separately analyzes the conversation turn and emits candidate structured signals for validator review.
- Persistence stage stores messages, analytics, safety outcomes, and approved memory/signal updates.

9.2 Minimum AI components

Component	Role
Intent classifier	Maps each user turn to approved intents and sub-intents.
Tone / stance detector	Identifies tone, activation, self-positioning, and conversational posture.
Safety detector	Finds self-harm, abuse, coercion, stalking, panic, crisis, or high-risk dependency language.
Structured extractor	Turns narrative content into canonical signal candidates.
Guidance generator	Produces the user-visible answer using evidence-banded context.
Helpfulness calibrator	Scores whether output felt useful and adjusts future prompt selection.
Cost router	Chooses lite vs standard vs deep path based on entitlement, complexity, and urgency.

9.3 Evidence band rules for user-visible outputs

Band	Internal meaning	Allowed language	Example
LOW	Sparse data, one-off mention, or contradictory evidence	use soft framing: 'might', 'it sounds like', 'one possibility'	'It sounds like delayed replies might be a trigger for you, but I would want more history before treating that as a settled pattern.'

Band	Internal meaning	Allowed language	Example
MEDIUM	Repeated cues or multi-source support but not stable enough for hard claims	use moderate framing: 'seems to', 'often', 'there is a pattern suggesting'	'There is a pattern suggesting that mixed signals often push you into overanalysis.'
HIGH	Strong multi-source support over time or validated by user confirmation	use clear but bounded framing: 'consistently', 'your data shows', 'you tend to'	'Across your assessment and recent reflections, you consistently protect yourself by getting quieter before conflict fully opens.'

9.4 Refusal and redirection classes

- Clinical diagnosis request: refuse diagnosis, offer reflective framing and encourage professional help if appropriate.
- Emergency or self-harm: stop normal mode, provide crisis-oriented guidance and escalation per policy.
- Requests to surveil or manipulate another person: refuse and redirect toward self-awareness and respectful communication.
- Legal / medical / financial claims outside scope: refuse specialized advice and state limits.
- Non-consented partner exposure: explain that the system cannot expose private partner data without consent.
- Harassment or abuse planning: refuse and escalate if required by safety policy.

9.5 Prompt architecture

- System prompt should encode product role, safety scope, evidence-band behavior, approved claim language, and no-diagnosis rules.
- Mode prompt should encode session mode goals, extraction recipe, and preferred answer structure.
- Context packet should provide only curated summaries and canonical fields, not uncontrolled raw dumps.
- Style layer should provide premium brand voice: calm, precise, elegant, non-clinical, and never melodramatic.
- Output schema should require fields such as response_text, follow_up_question, suggested_actions, evidence_band_used, extraction_candidates, and safety_outcome.

10. API Contracts, Service Boundaries, and Event Model

Quantum Mirror needs clean service boundaries so developers do not bury logic inside controllers or prompts.

10.1 Service map

- Entitlement Service - verifies subscription status and quotas.

- Context Assembly Service - composes the Quantum Mirror packet from Genesis OS stores.
- Conversation Service - owns session and message lifecycle.
- AI Orchestration Service - classifier, generator, extractor, and post-process pipeline.
- Signal Validator Service - checks canonical enums, thresholds, contradictions, and save policy.
- Memory Service - stores, deletes, and versions memory items.
- Moderation Service - detects risk, queues reviews, and triggers escalation.
- Analytics Service - logs product usage, model costs, helpfulness, and safety metrics.

10.2 Core request/response contracts

POST /qm/session/start

```
{
  "request": {
    "user_id": "uuid",
    "conversation_mode": "PAST_RELATIONSHIP",
    "entry_surface": "PROFILE",
    "partner_context_user_id": "uuid|null"
  },
  "response": {
    "qm_session_id": "uuid",
    "session_status": "OPEN",
    "entitlement_tier": "QM_PREMIUM",
    "remaining_message_quota": 120,
    "context_depth_tier": "STANDARD",
    "evidence_band_ceiling": "MEDIUM"
  }
}
```

POST /qm/message/send

```
{
  "request": {
    "qm_session_id": "uuid",
    "user_message": "text",
    "client_timestamp": "ISO8601"
  },
  "response": {
    "assistant_message_id": "uuid",
    "assistant_text": "text",
    "follow_up_question": "text|null",
    "suggested_actions": [
      "array"
    ],
    "evidence_band_used": "LOW|MEDIUM|HIGH",
    "safety_outcome": "NONE|FLAGGED|ESCALATED",
    "memory_save_prompts": [
```

```
      {
        "prompt_id": "uuid",
        "text": "Save this as a remembered pattern?"
      }
    ]
  }
}
```


POST /qm/memory/confirm


```
{
  "request": {
    "memory_prompt_id": "uuid",
    "decision": "CONFIRM|REJECT"
  },
  "response": {
    "status": "ok",
    "memory_item_id": "uuid|null"
  }
}
```


GET /qm/profile-summary


```
{
  "response": {
    "user_id": "uuid",
    "readiness_score": 74,
    "archetype_primary": "TRANSLATOR",
    "shadow_primary": "SELF_SABOTEUR",
    "top_trigger_tags": [
      "TRIGGER_DELAYED_RESPONSE"
    ],
    "active_theme_tags": [
      "THEME_EMOTIONAL_INVISIBILITY"
    ],
    "recent_state": "GUARDED_HOPEFUL"
  }
}
```


10.3 Event model

Field	Type	Description
event_id	uuid	Global unique id
event_type	enum	QM_SESSION_STARTED / QM_MESSAGE_SENT / QM_SIGNAL_PROPOSED / QM_MEMORY_CONFIRMED / QM_SAFETY_ESCALATED /

Field	Type	Description
		QM_RESPONSE_RATED / etc.
user_id	uuid	Actor / owner
subject_user_id	uuid	Optional second-party subject
qm_session_id	uuid	Optional session link
payload	json	Typed event body
version	varchar	Event schema version
created_at	timestamp	UTC

- Every AI-generated answer should log model path, prompt version, context packet version, evidence band, and token counts.
- Every extracted signal should log extractor version, source span ids, save policy outcome, and validator result.
- Every moderation action should log risk detector version, rule id, escalation target, and resolution status.
- Every subscription gate event should log entitlement snapshot and remaining quota at time of decision.

11. Moderation, Escalation, and Consent Architecture

11.1 Consent scopes

Scope code	Meaning	Allows save of	Does not allow
CONSENT_QM_EPHEMERAL	Use conversation only for current answer	session context	durable memory or profile extraction
CONSENT_QM_PERSONAL_MEMORY	Use and save personal memories and low-risk extracted signals	personal memory, stable preferences, history themes	couple/shared memory
CONSENT_QM_PROFILE_ENRICHMENT	Allow approved extracted signals to enrich Genesis OS personal profile	trigger tags, pattern tags, history entries	direct mutation of slow identity without rules
CONSENT_QM_COUPLE_SHARED	Allow selected shared summaries for couple mode when both users agree	shared session summaries, agreed tasks	unilateral private disclosures to partner
CONSENT_QM_TRAINING_AGGREGATE	Allow anonymized aggregate use for tuning and analytics	de-identified metrics	raw identifiable chat exports for model training without separate policy

11.2 Deletion and retention principles

- User must be able to view and delete durable Quantum Mirror memories and past relationship entries subject to legal/safety exceptions.
- Deleting a memory item should remove it from future context assembly immediately and queue downstream cache invalidation.
- Derived structured signals tied only to deleted content should be soft-deactivated or deleted according to retention class and policy.
- Moderation evidence may have longer retention than product memory when required for abuse handling, dispute resolution, or security.
- Aggregate analytics should be de-identified and excluded from user-visible memory controls while still honoring data governance commitments.
- Retention classes should be explicit: session transcripts, standard memories, sensitive memories, moderated evidence, billing records, and de-identified aggregates.

11.3 Moderation ladder

Risk class	Examples	AI behavior	System action	Human review
NONE	ordinary reflection, score questions	normal response	log only	no
LOW	mild distress, heartbreak, confusion	gentle reflective support	log sentiment and watch	no
MODERATE	panic language, repeated harassment concerns, coercive control hints	supportive but bounded, suggest safety-oriented next steps	open moderation case, flag session	possible
HIGH	self-harm ideation without imminent plan, threats, stalking, abuse disclosures with fear	interrupt normal flow, safety-first language	escalate immediately, provide resources, restrict some modes	yes
CRITICAL	imminent self-harm, active violence threat, child exploitation or comparable severe harm	stop normal assistance	critical escalation policy, emergency resources, account interventions where policy permits	yes immediate

11.4 Couple mode consent warnings

- Couple mode must never surface one partner's private Quantum Mirror session content to the other unless explicitly shared.
- Shared summaries should be generated from approved shared inputs, joint exercises, or transformed patterns that remove unilateral secrets.

- If one partner revokes consent, shared future analysis stops immediately and any cached joint context must be invalidated.
- All couple outputs should include a provenance class: SELF_REPORTED_BOTH, JOINT_ACTIVITY_DATA, SHARED_REFLECTION, or DERIVED_PATTERN_LOW_EXPOSURE.

12. Cost Control, Rate Limits, and Monetization Logic

Premium value does not excuse uncontrolled compute burn. Quantum Mirror must behave like a luxury service with a disciplined back office.

12.1 Cost levers

- Route simple questions to a smaller model path and reserve deeper model paths for complex pattern analysis.
- Cap the amount of historical context assembled per turn. Retrieve targeted summaries, not whole transcript novels.
- Use structured summaries and canonical signal lists instead of replaying every prior session.
- Limit automatic extraction frequency. Not every user turn needs an extraction job.
- Use monthly message/session quotas with soft warnings and optional top-up rules if desired.
- Use asynchronous enrichment jobs for non-urgent summarization and trend updates rather than deep synchronous generation every time.
- Log cost_estimate_usd per session and per subscription cohort to ensure margin remains positive.

12.2 Suggested entitlement matrix

Capability	Free preview	QM Premium	QM Couple	Notes
Ask unlimited basic questions	no	controlled	controlled shared	Keep hard cap hidden if possible; use fair-use language
Durable personal memory	no	yes	yes	Must honor consent and deletion
Past relationship guided intake	preview only	full	n/a initially	High-value premium differentiator
Message Studio	limited	yes	yes	Premium sticky feature
Couple reflection	no	no	yes	Later phase
Deep forecast interpretation	light	yes	yes	Only evidence-banded

13. Narrow-v1 Launch Path, QA, Analytics, and Rollout Gates

The right v1 is the smallest premium form that teaches the engine. Do not launch the whole empire at once.

13.1 Build v1 scope

- Single-user premium mode only. No couple-shared memory in v1.
- Conversation modes for v1: Score Deep Dive, Past Relationship Decode, Message Studio, and Pattern Check.
- Use personal profile summary, recent state, and selected memories. Do not unlock every Genesis OS subsystem on day one.
- Enable personal memory and extracted signal storage only for a limited set of low-risk canonical families: trigger tags, history themes, communication style tags, durable preferences, and growth signals.
- Moderation for self-harm, abuse, stalking, threats, and dependency language must be live before public rollout.
- Launch with user-visible memory review screen from the start. Trust compounds when users can see what the system thinks it knows.

13.2 Do later, not first

- Full couple reflection mode and shared summaries.
- High-impact automatic influence on readiness or compatibility without manual governance review.
- Open-ended always-on AI companion behavior that encourages dependency.
- Deep partner inference for users who only have unilateral data.
- Automatic Emotional Genome mutation directly from Quantum Mirror chat.
- Complex economy mechanics tied to every AI session.

13.3 QA checklist

- Can the assistant explain a score without sounding generic or overconfident?
- Do extracted signals always use canonical enums and valid time scopes?
- Can the user review, confirm, and delete saved memories cleanly?
- Does deletion invalidate future context assembly within acceptable latency?
- Do high-risk messages trigger correct moderation class and escalation?
- Does couple mode remain fully blocked when consent is absent or revoked?
- Are cost metrics logged and attributable to user, session, model path, and mode?
- Can engineering answer 'why did the system say this?' from logs alone?

13.4 Launch gates

- Gate 1 - Functional readiness: session lifecycle, billing gate, message send, basic response, logs.

- Gate 2 - Governance readiness: consent scopes, memory review controls, deletion handling, evidence-band enforcement.
- Gate 3 - Safety readiness: moderation ladder, escalation queue, audited policy tests, response interruption flows.
- Gate 4 - Quality readiness: helpfulness target, low genericity rate, low overconfidence complaints, acceptable latency and margin.
- Gate 5 - Learning readiness: dashboards for intent mix, extraction acceptance, memory confirms, churn correlation, and cost per active subscriber.

14. Step-by-Step Implementation Plan

Phase	What gets built
Phase 0 - Foundations	Lock canonical enums, create schema migrations, define entitlement matrix, and publish API contracts. No UI polish before truth model exists.
Phase 1 - Session backbone	Build qm_session, qm_message, entitlement checks, quotas, and analytics events. Support simple answer-only mode.
Phase 2 - Context assembly	Build read-only profile summary packet from Genesis OS. Include scores, top tags, recent state, and allowed memories.
Phase 3 - Guided generation	Implement prompt stack, evidence-band limiter, and response formatter for the first four conversation modes.
Phase 4 - Structured extraction	Add extractor pipeline, validator, qm_extracted_signal, and optional save-confirm UX.
Phase 5 - Memory and user controls	Add qm_memory_item, memory review UI, delete flow, retention jobs, and cache invalidation.
Phase 6 - Safety and moderation	Integrate risk classifier, moderation cases, escalation queue, and interrupted-response flows.
Phase 7 - Cost controls	Implement model router, context caps, quotas, session budgets, and cost dashboards.
Phase 8 - Beta and calibration	Run invite-only beta, analyze helpfulness, false memory saves, overclaiming, and cost. Tighten thresholds before public rollout.
Phase 9 - v1 public launch	Launch premium single-user Quantum Mirror with score deep dive, pattern check, message studio, and past relationship decode.
Phase 10 - v1.5 / v2 expansion	Only after stable evidence and moderation results: couple reflection, more extraction families, deeper forecast integration, and partner-shared surfaces.

Appendix A. Canonical Extraction Families

Family	Use	Examples
TRIGGER	What activates the user emotionally	TRIGGER_DELAYED_RESPONSE, TRIGGER_COLD_TONE, TRIGGER_MIXED_SIGNALS, TRIGGER_INCONSISTENCY, TRIGGER_CRITICISM, TRIGGER_WITHDRAWAL_AFTER_INTIMACY
HISTORY_THEME	Recurring narrative themes from past relationships	THEME_SELF_ERASURE, THEME_TRUST_FRACTURE, THEME_EMOTIONAL_INVISIBILITY, THEME_OVERSTAYING, THEME_RESCUER_ROLE, THEME_LOYALTY_TO_WOUND
COMMUNICATION_STYLE	How the user tends to communicate	COMM_SOFT_ENTRY, COMM_DIRECT, COMM_INDIRECT, COMM_OVEREXPLAIN, COMM_AVOIDANT, COMM_LOGICAL
RELATIONAL_NEED	What helps user feel secure and connected	NEED_REASSURANCE, NEED_CONSISTENCY, NEED_CLARITY, NEED_SOFTNESS, NEED_SPACE_WITH_REASSURANCE, NEED_ACTION_MATCHING_WORDS
BOUNDARY	What the user is no longer willing to accept	BOUNDARY_NO_MIXED_SIGNALS, BOUNDARY_NO_BEGGING, BOUNDARY_NO_SECRET_RELATIONSHIP, BOUNDARY_NO_NAME_CALLING
GROWTH_SIGNAL	Evidence of improved regulation or self-respect	GROWTH_PAUSED_BEFORE_REACTING, GROWTH_STATED_NEED_DIRECTLY, GROWTH_LEFT_EARLIER, GROWTH_REPAIRED_CLEANLY
SAFETY_SIGNAL	Risk cues requiring policy handling	SAFETY_STALKING_CONCERN, SAFETY_COERCIVE_CONTROL, SAFETY_SELF_HARM_IDEATION, SAFETY_THREAT_LANGUAGE

Appendix B. Sample Structured Signal Output


```
{
  "signal_id": "uuid",
  "user_id": "uuid",
  "source_qm_session_id": "uuid",
```

"source_qm_message_id": "uuid",
"signal_family": "TRIGGER",
"signal_code": "TRIGGER_DELAYED_RESPONSE",
"signal_value_type": "TAG",
"signal_value": {
"tag": "TRIGGER_DELAYED_RESPONSE",
"intensity": 0.72
},
"time_scope": "CURRENT",
"evidence_band": "MEDIUM",
"extractor_version": "qm_extract_v1.3",
"save_status": "PROPOSED",
"provenance_snippet": "late replies make my chest drop",
"created_at": "2026-03-19T18:20:00Z"
}

Appendix C. Sample Premium Response Template

1. Grounding sentence - reflect the user's situation without pretending certainty.

2. Pattern interpretation - connect current event to known tags or scores at the appropriate evidence band.

3. Practical guidance - give one to three actions, not a 40-piece self-help buffet.

4. Optional follow-up question - ask for the next most useful piece of context.

5. Optional save prompt - when a durable pattern likely emerged.

Example:

What you are describing sounds less like random overthinking and more like a pattern where delayed replies activate your body quickly. I am keeping that at a medium-confidence read because I have seen it in your assessment language and in two recent reflections, but not enough times yet to call it a settled trait. The useful move here is not to silence yourself or send five messages. It is to name what you need in one clean line and then watch whether their behavior matches it. When this happens, do you usually get louder, quieter, or more analytical inside?

Appendix D. Developer Build Checklist

- Publish canonical enum registry before coding client or service constants.
- Create migrations for all Quantum Mirror tables and event schemas.
- Implement entitlement enforcement and quota accounting.
- Build context packet assembler with strict field allowlist.
- Implement AI orchestration with mode-specific prompts and evidence limiter.
- Implement extractor + validator + signal persistence.
- Build memory review and delete UI.
- Integrate moderation ladder and escalation queue.
- Log token counts, cost estimates, latency, helpfulness, and overconfidence feedback.

- Run closed beta with transcript review, extraction audit, and safety audit before scale.